

PARSEREE MCU SUPPORT CIRCUIT - PIC18F57Q84-IPT 48-pin TQFP

Every resistor and capacitor shows its VALUE next to the reference designator. MPN / status / cost: mcu-bom.md
All pin numbers are the 48-pin TQFP column (DS40002213F Figure 2-7 / Table 3-2)

(1) U1 PIC18F57Q84-IPT : every connected pin

No AVDD / AVSS and no VCAP on this device, so ADC quality depends entirely on how clean 3V3_SYS is at pins 7 and 30: there is no analog supply to filter separately, and VREF is not a dedicated pin (it would cost RA2 / RA3). Unused I/O: drive low in firmware (DS40002213F 4.6) or tie to VSS through 1-10 kOhm.

(2) SUPPLY AND DECOUPLING : 0.1 uF per VDD pin, REQUIRED (DS40002213F 4.2.1)

Pins 7/6 and 30/31 sit on OPPOSITE CORNERS of the package, so these are two genuinely separate placements - one shared cap is not compliant. Each cap: ceramic, self-resonant at 200 MHz or above, same side of the board as the MCU, trace no longer than 6 mm, and power routed to the CAP FIRST and then on to the device pin.

(3) MCLR RESET NETWORK (DS40002213F Figure 4-1)

All three parts within 6 mm of pin 20 (4.3). R_mclr1 <= 10 kOhm and R_mclr2 <= 470 Ohm are datasheet limits, not suggestions: R_mclr2 caps the current C_mclr can dump into MCLR during an ESD/ESD5 breakdown. The Curiosity Nano fits 47 kOhm and no capacitor, which violates R1 <= 10 kOhm - follow the datasheet, not the dev board.

(4) ICSP PROGRAM + DEBUG HEADER - PICKIT 5 (Microchip PICkit pinout, Figures 48-1 / 48-2)

DS40002213F 4.4 is emphatic about what NOT to add: no pull-UPS, series diodes, or capacitors on ICSPCLK / ICSPDAT - they interfere with programmer communications. R_icap1/2 are ESD series only and must not exceed 100 Ohm; the 47 kOhm pull-DOWNS are what Microchip debug tooling expects. Nothing else may share RB6 / RB7.

THIS HEADER IS BOTH PROGRAMMER AND DEBUGGER, permanently, not a manufacturing future. The device offers In-Circuit Debug with three breakpoints over the SAME two pins (DS40002213F p. 8). MPLAB offers DEBUG (CONFIG7[5]) for a debug session and no board change is needed. JTAG is not an alternative - 39 states the PIC18Q84 JTAG module does not support programming. Leave JUP (CONFIG4[5]) at its erased 1: clearing it forces high-voltage VPP entry from then on, and only HV programming can ever set it back.

(5) DEBUG CONSOLE - USB-C to CP2102N-A02-GQFN24, CDC virtual COM port (console half of the Curiosity Nano debugger, DS50003011 3.1.2)

D_dbg (USBLC6-25C6) pin hookup - wire BOTH pins of each I/O pair:
pin 1 IO1 +
pin 6 IO1 +
pin 3 IO2 -
pin 4 IO2 -
pin 5 VBUS
pin 2 GND

US runs from 3V3_SYS, NOT from VBUS. This is CP2102N Figure 2.3 (internal 5 V regulator unused, so VREGIN is tied to VDD) plus Figure 2.6 (self-powered USB, VBUS reaching the SENSE PIN ONLY). The bus-powered Figure 2.5 ties VBUS to VREGIN instead - that is the wiring to avoid, because it would let a laptop pin the bridge while the vehicle rail is dead and drive DBG_RX into an unpowered MCU. The back-power failure mode from bench-wiring.md is designed out rather than patched. VBUS only ever tells US that a host is attached.

THE CROSS IS ALREADY DONE: US TXD (21) drives the PIC RX (RF1) and US RXD (20) listens to the PIC TX (RF0), exactly as the Nano wires CDC_TX to the target UART RX (DS50003011 Figure 3-1). Do not cross it a second time. Both directions must be populated: RF1 is how a server URL will later be written into the 1 KB Data EEPROM, so an output-only console would foreclose it. No DTR/RTS auto-reset circuit - reset belongs to J_icsp, and an RC on MCLR outside the sheet-3 network would light the PICkit for the pin. Two SEPARATE 5.1 kOhm CC resistors are required, and D+&D- must reach BOTH A6/A7 and B6/B7 for either cable orientation to work. Route the pair at 90 Ohm differential. No series resistors on D+&D-: the CP2102N integrates the transceiver, its matching, and the pull-ups. Programming is NOT here - the PIC18F57Q84 has no USB peripheral, so US is the virtual COM port only. Program and debug are on J_icsp with a PICkit 5.

(6) STATUS LED - RF3, pin 39, active low

HFINTOSC at 64 MHz, Y_n, C_osc1, C_osc2 and Y_sosc are all quantity ZERO, which leaves OSC1 (32) and OSC2 (33) free as general I/O. CONFIG1 MUST be programmed with RSTOSC = 000: an erased part comes up with RSTOSC = 111, meaning "external clock", and a crystal-less board will simply never run.